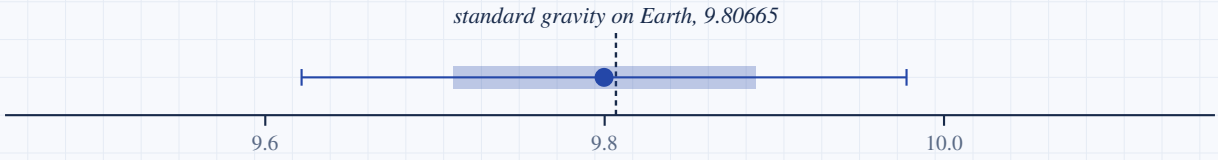
Rigid pendulum — rod with wooden block

Measured from a video (shmvideo5pendulum.mp4): SAM 3.1 tracked the motion and Muse Wonder (muse-wonder-1.0) worked out the physics.

Gravitational acceleration g

blue bar ±1σ, whiskers ±2σ

9.800 ± 0.089 m/s²



0.071% below standard gravity on Earth (9.80665 m/s²): inside the error bar.

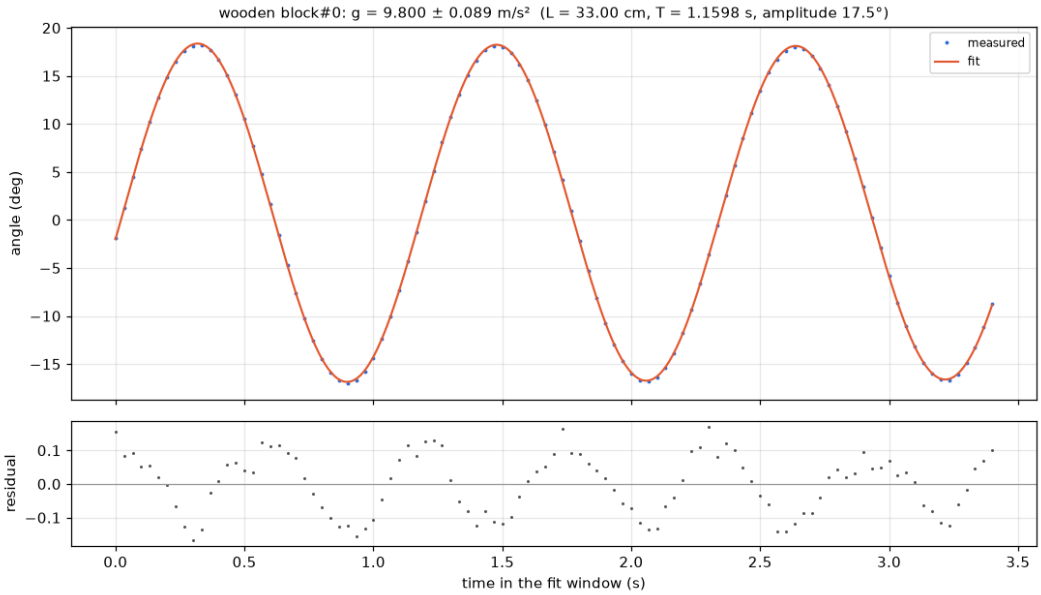
Measured g = 9.80 ± 0.09 m/s² from the pendulum period and the given 33 cm length, within 0.1% of the standard 9.80665 m/s² and consistent within the uncertainty.

Oscillation period (at observed amplitude)	Small-angle period	Swing amplitude
1.15976 ± 0.00016 s	1.153 s	17.5 deg

The law

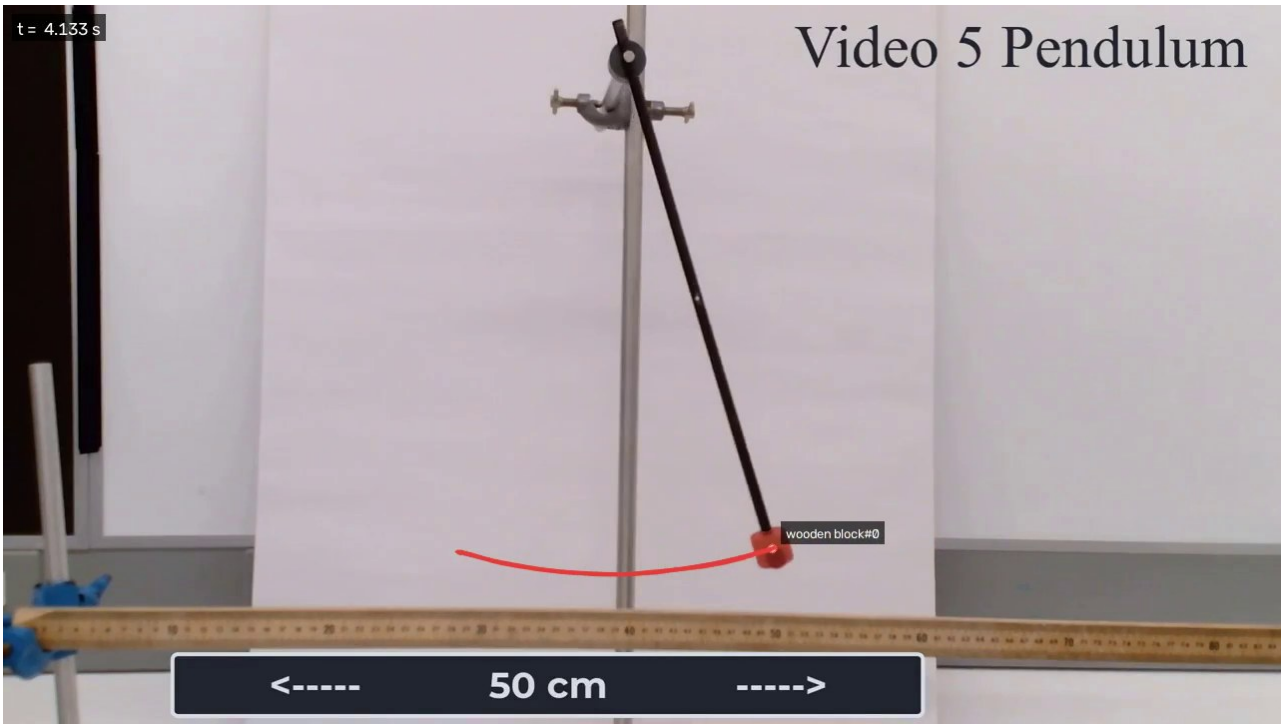
θ(t) = θ0·e^(−γt)·cos(2πt/T + φ), with T ≈ 2π√(L/g) corrected for finite amplitude; L is the given effective length (pivot to centre of mass).

The fit



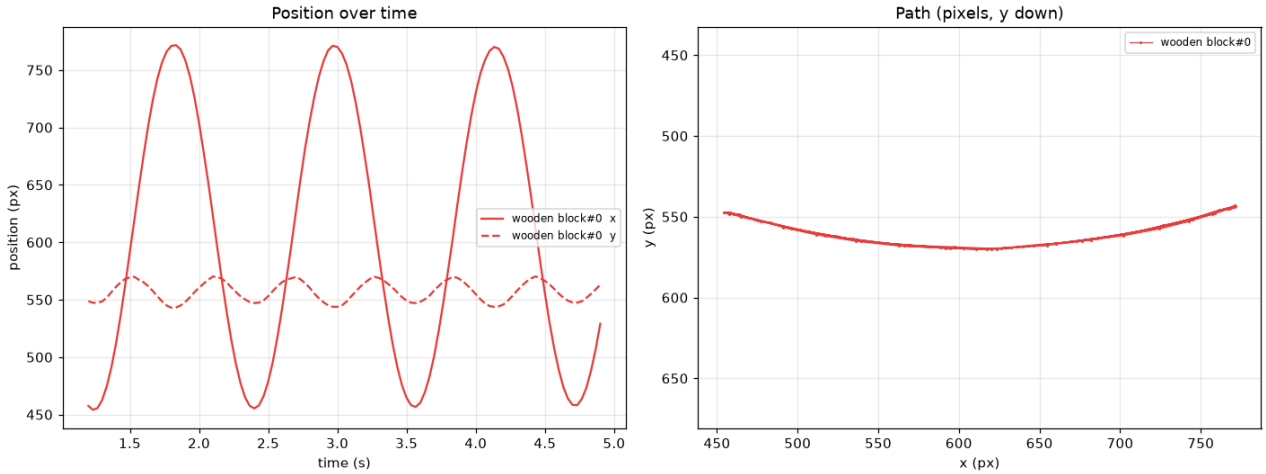
Tracked data (dots) with the fitted law (line); the residuals underneath show what the law does not explain.

What SAM 3.1 tracked



SAM 3.1's mask on “wooden block” at t = 4.13 s, with its recent path.

Prompt	“wooden block”
Tracked	wooden block: 112 of 148 frames (76%), no gaps
Clip	4.9 s at 30 fps, 1280×720 px
Length scale	None in the video, so positions are in pixels; lengths come from the notes.



Position of every tracked object over time, and its path in the picture.

All results

Quantity	Value	Unit	Note
Gravitational acceleration g	9.800 ± 0.089	m/s²	±1 standard error; dominated by the 0.3 cm uncertainty in the given 33.0 cm length (≈0.9%), since the period is far more precise. Compared with 9.80665 m/s², difference −0.07%.
Oscillation period (at observed amplitude)	1.15976 ± 0.00016	s	±1 standard error; scale-free timing measurement, 103 points in 1.5–4.9 s window.
Small-angle period	1.153	s	Amplitude-corrected period reported by the pendulum fit; shorter than the observed period as expected at finite amplitude.
Swing amplitude	17.5	deg	Moderate amplitude, so the finite-amplitude period correction matters at the ~0.6% level.
Arc radius of tracked block point	523.4	px	Cross-check of pivot location and circular arc only (arc RMS 0.42 px). This is the pivot-to-tracked-point distance, not the effective pendulum length, so it was NOT equated to 33 cm and no px-to-cm scale was set.
Damping rate	0.006083	1/s	Very light damping; quality factor ≈445 from the pendulum fit. Only ~3 cycles were recorded, so damping is poorly constrained compared with the period.

What this means

A rod with a wooden block on the end was swung like a pendulum and its block was tracked through about three swings. Its angle repeats every 1.1598 seconds, a timing that needs no distance scale. Combining that period with the lab's 33 cm effective length in the pendulum law gives g = 9.80 ± 0.09 m/s², essentially the accepted Earth value. The uncertainty is dominated by the length measurement, not the timing, and the block's pixel arc radius was used only to confirm the circular motion and pivot, not as the pendulum length.

Laws compared

Model	BIC	rms	Verdict
Nonlinear pendulum fit (fit_model, L = 33.0 cm given)			Accepted: gives g = 9.80 ± 0.09 m/s², angle RMS 0.088°, arc RMS 0.42 px, no warnings. This is the physical model including finite amplitude.
Damped simple-harmonic θ = A·e^(−γt)·cos(ωt+p)	-888.9	0.0122	Preferred of the two cosine approximations, but leaves systematic residuals because a 17.5° pendulum is not a pure sinusoid.
Undamped simple-harmonic θ = A·cos(ωt+p)	-880.9	0.013	Worse: BIC higher by 7.9 (modest, not decisive >10) and RMS higher; damping is small but the damped form fits slightly better. Both cosine forms are approximations to the nonlinear pendulum.

BIC: lower is better; a difference above 10 is decisive.

Data quality

- Track 'wooden block#0': 112 points, 1.2–4.9 s, no gaps (largest gap 0 frames); 103 points used from 0.2 s after release.
- Motion follows a circular arc very cleanly: arc RMS 0.42 px about the fitted pivot above the path, spanning 35.3°.
- Only about 3.1 cycles were recorded, so the period is precise but damping/Q is weakly constrained.

- No pixel scale was available and none was needed for g: timing plus the given length determine g. The tracked point's pixel arc radius must not be identified with the effective length of a rigid pendulum.
- Residuals of simple cosine fits are highly correlated (Durbin–Watson ≈0.007), consistent with finite-amplitude anharmonicity rather than random tracking noise.

Next steps

- Measure the effective length more precisely (or determine it independently for the rigid rod+block), since it dominates the g uncertainty.
- Record many more swings to pin down the small damping and improve the period further.
- Repeat at a smaller amplitude to reduce reliance on the finite-amplitude correction, and at other lengths to check $T \propto \sqrt{L}$.
- Calibrate the pixel scale with the ruler visible in the video if the pivot-to-tracked-point distance in cm is also wanted as a geometry cross-check.

How Galileo worked it out

Muse Wonder (muse-wonder-1.0) read the tracks and the notes, then called its own fitting tools. Every number in this report comes from those tools.

Step 1. Rigid pendulum (rod + block) swinging on an arc — period is scale-free, so I'll fit it with the given 33 cm length, not the block's arc radius.

fit_model(model=pendulum, track=wooden block#0, length_cm=33.0)
wooden block#0: g = 9.800 ± 0.089 m/s² (L = 33.00 cm, T = 1.1598 s, amplitude 17.5°)

Step 2. Pendulum fit gives g = 9.80 m/s² with a clean arc. Now I'll test the damped-oscillation law against an undamped alternative using BIC.

fit_equation(series=theta, expression=A*exp(-gamma*t)*cos(w*t + p), track=wooden block#0, start=1.5, end=4.9, initial={'A': 0.3, 'gamma': 0.01, 'w': 5.42, 'p': 0.0})
rms 0.01222, BIC -888.9, R² 0.99683

fit_equation(series=theta, expression=A*cos(w*t + p), track=wooden block#0, start=1.5, end=4.9, initial={'A': 0.3, 'w': 5.42, 'p': 0.0})
rms 0.01299, BIC -880.9, R² 0.99642

submit_report()
report received

Method and provenance

Video	shmvideo5pendulum.mp4
Notes given	Rigid pendulum: a rod with a wooden block on the end. 33 cm from the pivot to the centre of mass (given by the lab).
Tracking	SAM 3.1 on the Meta Model API: one call per prompt, masks for every frame
Analysis	Muse Wonder (muse-wonder-1.0): 3 model calls, 4 tool calls, 53 s
Tokens	18,067 input (5,236 cached), 5,100 output, about $0.15
Files	runs/shmvideo5pendulum-1103-142726: tracks.csv, overlay.mp4, report.json, agent_transcript.md and the fit plots